

Answer Evaluation using TrOCR and Rubric-Guided Transformer-based NLP

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Abstract—Manual grading of student answers is time-consuming and subjective, particularly for handwritten answer submissions. There is an increasing need for an automated answer evaluation system to evaluate student answers, which will improve accuracy and reduce the evaluation time. This paper presents a novel rubric-guided automatic evaluation system which evaluates both typed and handwritten responses. The students' handwritten answers are processed by using Transformer-based Optical Character Recognition (TrOCR) supported by Convolutional Neural Networks (CNN) for text extraction, while the typed answers are directly analyzed using the Transformer-based model, BERT. The system evaluates student responses based on their answer similarity to reference answers, grammatical correctness, and predefined rubric-based scoring method to generate marks and feedback. Evaluation on 400 student answers achieved Spearman and Pearson correlation coefficients of 0.9351 and 0.9419, respectively, demonstrating strong alignment with the human evaluation method. The results show that the proposed system provides consistent and reliable grading while reducing evaluator workload and grading time.

Keywords—Automatic Answer Evaluation, Optical Character Recognition, Transformer Models, Rubric-based Scoring, Semantic Similarity, Natural Language Processing.

I. INTRODUCTION

A rapid increase in the use of digital learning platforms and online examinations has increased the need for student assessment methods that are both fast and accurate. Multiple-choice questions can be evaluated automatically with high accuracy, but the evaluation of descriptive answers still relies heavily on manual grading. This manual process is time-consuming and may lead to inconsistencies, particularly when a large number of student responses must be evaluated within limited time. Frequent assessments and many students' responses further slow down the grading process. Handwritten responses vary significantly in writing style, character shape and formatting, making them difficult to process using conventional text-based evaluation techniques.

The advancements in artificial intelligence, particularly in deep learning and natural language processing (NLP), have enabled systems to analyse and understand human language more effectively than traditional approaches. Transformer-based models such as BERT are capable of capturing contextual and semantic meaning from text. By using these technologies, automated answer evaluation systems can be developed to assess both typed and handwritten student responses more accurately and consistently.

A rubric-guided automatic answer evaluation system is implemented and evaluated using Transformer-based Optical Character Recognition (TrOCR) and a transformer-based language model, BERT. The system extracts text from

handwritten answer images and processes both typed and handwritten responses through a unified evaluation pipeline. Student answers are assessed based on semantic similarity with reference answers, grammatical correctness using LanguageTool, and predefined rubric-based scoring criteria provided by evaluators. The system generates qualitative feedback to enhance students' performance. The objectives of the implemented system are to reduce evaluator workload, improve grading consistency, and provide timely feedback to students, thereby enhancing the overall assessment process.

II. RELATED WORK

Automatic evaluation of descriptive answers has been a key research area in educational technology. Early studies [1] primarily relied on rule-based and keyword-matching approaches for evaluating short answers. Although these methods reduced the manual workload of evaluators, they failed to capture the true semantic meaning of student responses.

Tsihrintzis and Jain [2] explored traditional automated assessment frameworks that focused more on surface-level textual features and failed to understand the actual intent and content of student answers.

With advancements in natural language processing (NLP), Semantic Similarity-based methods for automatic grading have gained increasing attention. Yang et al. [3] proposed an automatic short answer grading system using contextual embeddings to measure semantic similarity between student responses and reference answers. While this approach achieved improved accuracy compared to keyword-based methods, it did not support handwritten answer evaluation.

Bittencourt et al. [4] combined neural networks and NLP techniques with educational assessments to improve grading reliability. However most existing systems focus primarily on numerical score prediction rather than a structured rubric-based scoring and providing instant, detailed feedback for students' answers.

Additional challenges in evaluating handwritten student answers are due to variations in writing styles, noise in scanned images, and OCR recognition errors. Memon et al. [5] employed CNN-based OCR techniques to process handwritten answers and showed improved text recognition accuracy.

Agarwal et al. [6] demonstrated the effectiveness of deep learning-based frameworks used for handwritten text

recognition and showed the importance of image preprocessing and feature extraction in evaluation.

Mahadevkar et al. [7] proposed an NLP and machine learning-based model for evaluating student answers by assessing answer relevance, but the approach was limited to typed responses. Geetha et al. [8] introduced a structured grading framework that also focused only on typed answers.

Providing feedback to students is another important aspect of effective assessment. Studies on automatic feedback generation [9] demonstrated how NLP techniques can help in identifying the strengths and weaknesses in student answers and support learning.

Kayser [10] and Looi et al. [11] showed the use of AI-driven assessment tools in large-scale educational environments. Chakraborty et al. [12] proposed an AI-based assessment framework that integrates multiple NLP models to improve accuracy. But the system lacked a structured OCR pipeline and support for multimodal inputs.

The study by Wang et al. [13] explores the use of BERT for automated essay scoring by capturing the deep contextual relationships within the text. The sentence-level and document-level features considered for scoring show the multi-scale representation of the system. Lubis et al. [14] conducted a study focused on word embeddings using vector representations to capture semantic similarity between student answers and model answers.

Nie et al. [15] proposed a model that combines SBERT embeddings with Long Short-Term Memory (LSTM) networks and attention mechanisms. SBERT is used to generate sentence embeddings, which are then processed by an LSTM network to capture sequential dependencies in the text. The attention mechanism enhances the model by focusing on key parts of answer that contribute to scoring. This research reinforces the effectiveness of using transformer-based models in an automatic answer evaluation system.

Most existing approaches focus on typed responses and do not simultaneously include handwritten answer evaluation, structured rubric-based scoring, and feedback generation. The implemented system integrates OCR-based handwritten text extraction, transformer-based semantic evaluation, rubric-guided scoring, and automated feedback generation to provide a comprehensive and consistent evaluation framework for both typed and handwritten student responses.

III. METHODOLOGY

The implemented system is designed to evaluate both typed and handwritten student answers using Deep Learning (DL) and Natural Language Processing (NLP) techniques. A pretrained transformer-based language model, BERT, is employed for the semantic analysis of student responses. The Mohler dataset is utilized for semantic similarity calibration and validation, as it contains Computer Science questions with multiple student responses graded with human-assigned scores ranging from 0 to 5, making it suitable for evaluating automatic answer assessment performance. The dataset was

split into training and validation sets in an 85:15 ratio. For stable convergence, a warmup of 100 steps was employed at the beginning of training to gradually increase the learning rate. Hyperparameters were selected based on empirical experimentation for fine-tuning the pre-trained transformer-based BERT model. The model was trained for 3 epochs with a batch size of 8. A sequence length of 512 tokens was used to ensure that long answers could be effectively captured during evaluation. The model was optimized using an AdamW optimizer with a learning rate of $2e-5$.

The system architecture integrates Transformer-based Optical Character Recognition (TrOCR), Convolutional Neural Networks (CNNs), and the BERT model to ensure accurate and consistent evaluation of student answers. The system understands the semantic meaning, rubric-guided weighted scoring, and automated feedback generation. Typed answers are directly processed through the NLP pipeline, while handwritten answers undergo an additional text extraction stage before evaluation.

A. Dataset and Experimental Setup

To evaluate the effectiveness of the implemented system, experiments were conducted on a dataset consisting of 400 student answers. Each answer was graded on a scale from 0 to 2. Handwritten responses were provided as scanned images, while typed answers were collected in textual format. The answers consist approximately 10 to 30 words and are related to computer science topics. The evaluation was performed to assess grading accuracy and consistency of system-generated scores compared with human-assigned scores, as shown in Fig. 5.2.

IV. SYSTEM ARCHITECTURE

Fig. 4.1 illustrates the architecture of the implemented automatic answer evaluation system. The framework integrates handwritten text extraction, semantic evaluation, rubric-based scoring, and feedback generation to assess both typed and handwritten student responses in a unified pipeline.

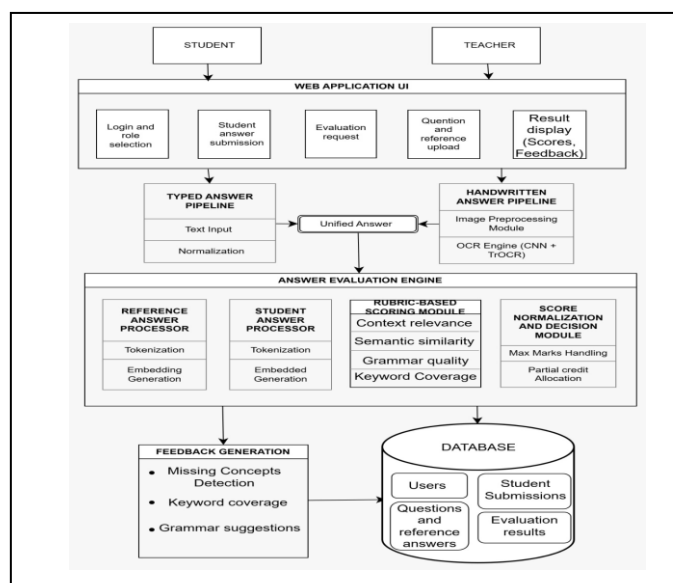


Fig. 4.1 Architecture diagram of the proposed system

A. Preprocessing and Text Extraction

Handwritten answer images often contain noise, variations in lighting conditions, skewness, and variations in student handwriting styles. To enhance text clarity and improve recognition accuracy, preprocessing steps like grayscale conversion, noise removal, contrast enhancement, binarization, resizing, and skew correction are applied. Outliers such as extremely short or irrelevant answers are filtered using length-based thresholds.

Following preprocessing, handwritten images are processed using a Transformer-based Optical Character Recognition (TrOCR) supported by a convolutional neural network (CNN). Transformer-based architecture of TrOCR enables effective modelling of long-range dependencies by using self-attention mechanism to capture relationships between tokens in the answer image. Unlike traditional OCR systems like CNN, RNN based approaches, TrOCR has Vision Transformer (ViT) encoder and transformer decoder, which capture variations in handwriting styles, contextual dependencies between characters better.

The CNN component extracts spatial features such as stroke patterns, curves, and character shapes through convolution and pooling layers, enabling robust recognition of varied handwriting styles. Once text is extracted, both typed and handwritten responses are converted into machine-readable format and passed to the textual preprocessing stage. This stage includes tokenization, lowercasing, removal of unnecessary symbols, whitespace normalization, and lemmatization to ensure uniformity and reduce textual noise before further analysis.

B. Semantic Similarity Scoring

The semantic similarity between the reference answer and the student answer is computed by using Sentence BERT (SBERT), which is a transformer-based architecture that produces dense vector representations for entire sentences, making it suitable for similarity computation. The reference answer and the student answer are converted into embedding vectors and the semantic similarity between these vectors is computed using cosine similarity as shown in (1):

$$\text{Similarity} = \frac{A \cdot B}{\|A\| \times \|B\|} \quad (1)$$

where A represents the embedding of the student's answer and B represents the embedding of the reference answer. The similarity score range from 0 to 1, higher similarity values indicate stronger semantic alignment with the reference answer.

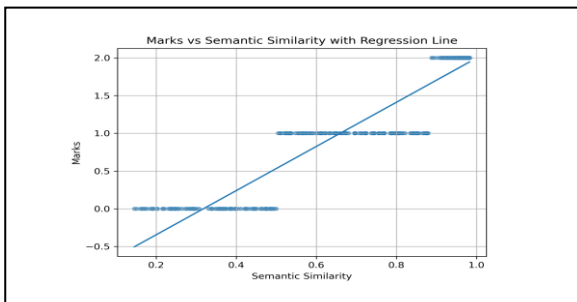


Fig. 4.2 Semantic Similarity of student's responses with their marks.

Fig. 4.2 illustrates the relationship between semantic similarity scores and the marks for student responses.

C. Contextual Similarity Scoring

In addition to semantic similarity, contextual relevance is evaluated to ensure that student responses align with the scope of the question. Unlike SBERT, which encodes sentences independently, a transformer-based language model, BERT, which relies on attention mechanism is used to generate contextual embeddings for both student answers and reference answers provided by evaluators. The cross-encoder, BERT will process the reference answer and student answer together as a single input sequence. The input is then passed through the system to capture the fine interactions between tokens of both student and reference answer.

The self-attention computes interactions between the tokens in a sequence, enabling the model to capture short-range and long-range dependencies irrespective of positions, where key concepts of the answer may appear in different parts of the response. The multi-head attention learns different types of relationships simultaneously, which helps in understanding the semantic meaning of student answers by considering contextual relationships, syntax, and sentence structure. The relevance score is computed based on the contextual alignment between student answer and the expected answer. The output logits are transformed into normalized score using a sigmoid function as shown in (2):

$$S_{context} = \sigma(\text{logits}) \quad (2)$$

Where σ denotes sigmoid function. The score ranges from 0 to 1 and it demonstrates how well the student response matches with the reference answer in terms of context, key concepts.

D. Grammar Assessment

The grammatical quality of student answers is evaluated using LanguageTool. The tool analyzes sentence structure, spelling errors, punctuation, and fluency. The grammar score is computed by normalizing the number of detected errors with respect to the total number of words in the answer as shown in equation (3):

$$S_{grammar} = 1 - \frac{\text{Number of Errors detected}}{\text{Total words in a answer}} \quad (3)$$

The score range between 0 and 1, higher score represents better grammatical correctness. This ensures that responses are assessed not only for content accuracy but also for language quality.

E. Rubric-based Scoring

The final score for a student answer is calculated using a rubric-based weighted scoring mechanism that combines contextual relevance, semantic similarity, and grammatical correctness. Each component contributes to the final score of the answer based on predefined weights. The weights assigned to each component are configurable and can be adjusted by evaluators based on assessment requirements. The final score (4) is computed as:

$$\text{Final Score} = w_c \cdot S_{context} + w_s \cdot S_{semantic} + w_g \cdot S_{grammar} \quad (4)$$

where $S_{context}$, $S_{semantic}$ and $S_{grammar}$ represents the contextual relevance score, semantic similarity score, and the grammatical score of the answer. The w_c , w_s and w_g are the weights which determine the contribution of each component to the final score of the answer. [For example, if $S_{context} = 0.96$, $S_{semantic} = 0.98$ and $S_{grammar} = 1$, and if the $w_c = 0.4$, $w_s = 0.4$ and $w_g = 0.2$, then the final score will be $(0.4 * 0.96) + (0.4 * 0.98) + (0.2 * 1) = 0.97$]. The weight of 0.4 assigned for both context and semantic similarity score is higher as these component scores directly reflect the student's knowledge, while 0.2 for grammar contributes to language clarity. This weighted scoring approach ensures balanced evaluation by considering both content accuracy and language quality.

The final score is normalized between 0 and 1 and then scaled to grading range (0 to 2) for final evaluation result. Different thresholds were applied for interpreting the marks of student response, for typed answers if final Score < 0.5 then mark = 0, if final Score lies between 0.5 to 0.88 then mark = 1, if score ≥ 0.88 then mark = 2. For handwritten answers, if final Score < 0.4 then mark = 0, if final Score lies between 0.4 to 0.8 then mark = 1, if score ≥ 0.8 then mark = 2. These threshold-based approach ensure fairness by accommodating OCR noises in handwritten text recognition process.

F. Feedback Generation

In addition to numerical scoring, the system generates qualitative feedback for students based on their evaluation results. The feedback points out the missing concepts, grammatical errors, and areas requiring improvement. Feedback is generated by comparing student responses with reference answers, enhancing the future performance of students.

The system is implemented using a Python-based deep learning and NLP framework. BERT is utilized for semantic and contextual evaluation, TrOCR combined with CNNs performs handwritten text recognition, and LanguageTool is used for grammar analysis.

V. RESULTS AND DISCUSSION

The performance of the implemented automatic answer evaluation system was evaluated using a dataset of 400 student answers, comprising both typed and handwritten responses.

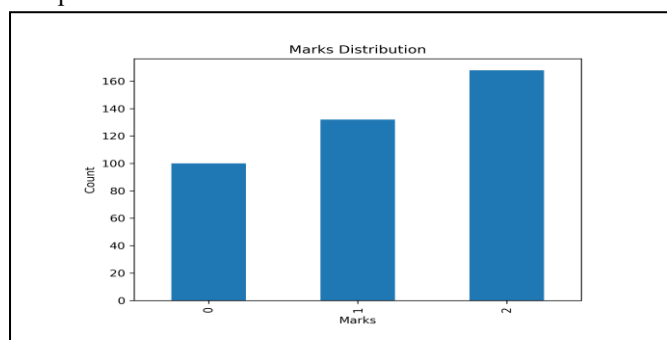


Fig. 5.1 Mark distribution of answers.

Fig. 5.1 shows the distribution of marks assigned by the automated evaluation system, indicating a balanced spread of

scores across the evaluation scale. A comparative analysis between automated scores and human-assigned scores was done using statistical and correlation-based evaluation metrics. The Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) values of 0.3377 and 0.4472, respectively, indicate a low average deviation between system-generated scores and human evaluation results. The coefficient of determination (R^2) value of 0.6880 demonstrates that a substantial proportion of the variance in human grading is captured by the automated scoring model. The Inter-rater Agreement was calculated by using Cohen's Kappa coefficient, a substantial agreement score of 0.6899 indicates that human grading was consistent and reliable. The high Pearson and Spearman correlation coefficients of 0.9419 and 0.9351 confirm a strong positive relationship between automated and human scores, indicating consistent grading.

The transformer-based semantic similarity module contributes significantly to evaluation accuracy by effectively capturing the meaning of student responses, even when different wording is used. In addition, the OCR-based handwritten text recognition module enables reliable text extraction from handwritten answers, making automated evaluation feasible for non-typed responses. TrOCR-based system got Character Error Rate (CER) of 0.1514 and Word Error Rate (WER) of 0.2655. The higher WER is due to minor spelling variations and character-level distortions in handwritten text images, even small errors can result in complete word mismatches. The moderate CER shows that most characters are correctly recognized, preserving the overall semantic meaning of the text. The semantic evaluation models, SBERT and cross-encoder BERT, help mitigate the impact of OCR errors by focusing on contextual and semantic similarity rather than exact lexical matching, ensuring better evaluation. The contextual relevance score component ensures that student answers remain aligned with the question requirements, while the grammar assessment module evaluates language quality, resulting in a more comprehensive assessment process.

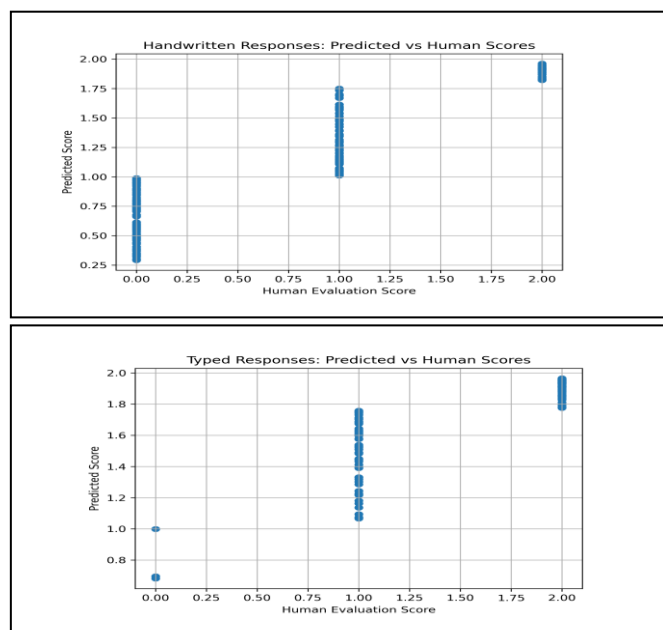


Fig. 5.2 Comparison of human scores with automated answer score prediction of both typed and handwritten responses.

Fig. 5.2 illustrates a strong linear relationship between automated and human scores, further validating the reliability of the proposed evaluation framework.

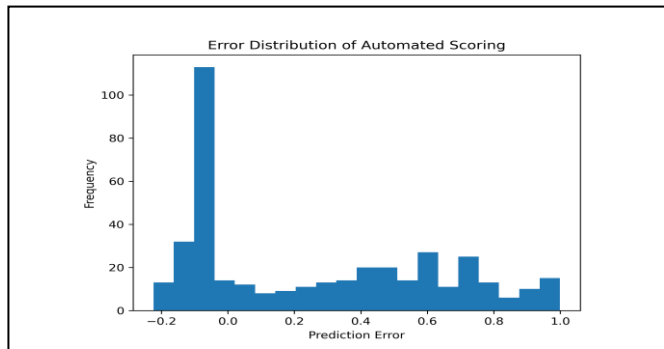


Fig. 5.3. Distribution of error in automated grading results.

Fig. 5.3 shows the error distribution, where most prediction errors fall within a narrow range, indicating stability in the scoring method.

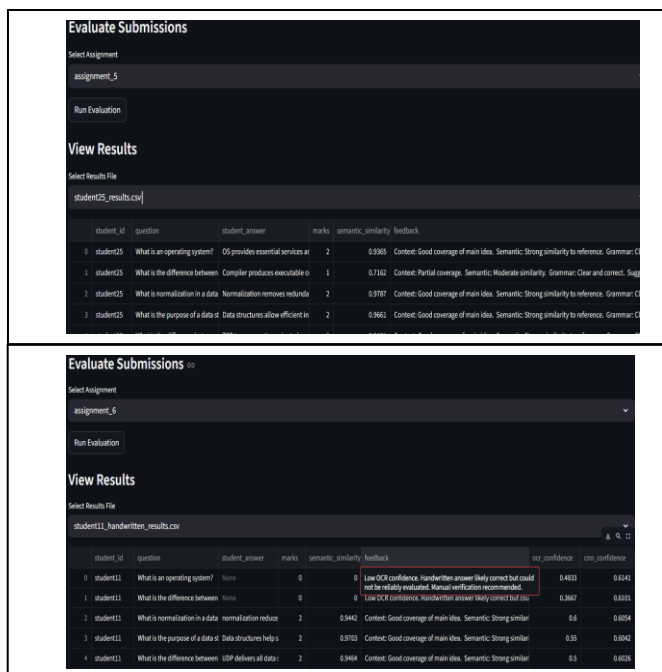


Fig. 5.4. Student result output of both typed and handwritten answers.

Fig. 5.4 presents sample evaluation outputs for both typed and handwritten answers. The computational efficiency was evaluated by measuring execution time for both typed and handwritten answers. For typed responses, the average processing time is approximately 2.22 seconds per answer, with semantic evaluation stage takes approximately 0.098 seconds. For handwritten responses, the OCR stage takes an average of 4.60 seconds, resulting in total processing time of approximately 6.51 seconds per answer. The higher processing time for handwritten responses is due to computational complexity of OCR stage. Overall, these results demonstrate that the implemented system can evaluate descriptive student responses with accuracy and consistency, while significantly reducing manual grading effort.

VI. CONCLUSION AND FUTURE WORK

A key challenge in handwritten answer evaluation is text recognition, as the performance of TrOCR module is influenced by handwriting quality. The CER and WER values increase in case of extremely poor or more different handwriting, which leads to error propagation. These inaccuracies may affect the reliability in scoring when key answer terms are not recognized properly. The proposed approach relies on limited and computer science domain-specific dataset, it lacks the diversity required to understand different student responses across various subjects, writing styles. The system does not evaluate answers that include diagrams and equations. Long descriptive answers require these, but these are not fully captured by sentence-level similarity metrics. To overcome these limitations, the future work will focus on fine-tuning TrOCR model on domain-specific handwritten datasets to improve recognition accuracy, and reducing CER and WER value to minimize error propagation. Incorporating post-OCR correction techniques using language models will further enhance transcription quality of the answer. Expanding the dataset containing diverse subject, answer styles will improve the generalization and robustness of the system. Developing the system to process diagrams, mathematical equation, symbols, and assessing voice-based responses would enable a comprehensive assessment framework. The proposed system included the deep learning and rubric-guided scoring mechanism with feedback generation for assessment of student answers, addressing the limitations will enhance the system to be robust and effective for answer grading. This paper presented a rubric-guided transformer-based automatic student answer evaluation system capable of assessing both typed and handwritten student responses using deep learning and natural language processing techniques. By integrating Transformer-based Optical Character Recognition (TrOCR) with CNN-supported handwritten text recognition and transformer-based semantic analysis, the system evaluates answer relevance, semantic accuracy, and grammatical quality. The rubric-based weighted scoring mechanism ensures balanced and consistent grading, which is closely aligned with human evaluation methods. In addition to generating numerical scores, the system provides qualitative feedback to support student learning and performance improvement. The experimental results demonstrate that the implemented system effectively reduces manual grading time while maintaining accuracy and consistency. The framework offers a scalable solution for evaluating descriptive answers in both online and offline educational environments.

REFERENCES

- [1] Z. Saidane, *Image and Video Text Recognition Using Convolutional Neural Networks*. LAP Lambert Academic Publishing, 2011.
- [2] G. A. Tsahrintzis and L. C. Jain, *Machine Learning Paradigms: Advances in Deep Learning-based Technological Applications*. Springer Nature, 2020.
- [3] R. Yang, J. Cao, Z. Wen, and J. Shen, "Automated post scoring: evaluating posts with topics and quoted posts in online forum," *World Wide Web*, vol. 25, no. 3, pp. 1197–1221, Mar. 2022, doi: 10.1007/s11280-022-01005-6.
- [4] I. I. Bittencourt, M. Cukurova, K. Muldner, R. Luckin, and E. Millán, *Artificial Intelligence in Education: 21st International Conference, AIED 2020, Ifrane, Morocco, July 6–10, 2020, Proceedings, Part II*. Springer Nature, 2020.

- [5] J. Memon, M. Sami, R. A. Khan, and M. Uddin, "Handwritten optical character recognition (OCR): A comprehensive systematic literature review (SLR)," *IEEE Access*, vol. 8, pp. 142642–142668, 2020, doi: 10.1109/access.2020.3012542.
- [6] V. Agrawal, J. Jagtap, S. Patil, and K. Kotecha, "Performance analysis of hybrid deep learning framework using a vision transformer and convolutional neural network for handwritten digit recognition," *MethodsX*, vol. 12, p. 102554, Jun. 2024, doi: 10.1016/j.mex.2024.102554. *MethodsX*, vol. 12, p. 102554, Jun. 2024, doi: 10.1016/j.mex.2024.102554.
- [7] S. Mahadevkar, S. Patil, and K. Kotecha, "Enhancement of handwritten text recognition using AI-based hybrid approach," *MethodsX*, vol. 12, p. 102654, Jun. 2024, doi: 10.1016/j.mex.2024.102654.
- [8] "Design of automated model for inspecting and evaluating handwritten answer scripts: A pedagogical approach with NLP and deep learning," *Alexandria Engineering Journal*, vol. 108, pp. 764–788, Dec. 2024, doi: 10.1016/j.aej.2024.08.067.
- [9] "Automatic Feedback Generation In NLP: A Systematic Review." Available: <https://doi.org/10.1109/ICAC2N63387.2024.10895143>. [Accessed: Jan. 13, 2026]
- [10] C. Grévisse and F. Kayser, "Lessons learned from conducting assessments in radioanatomy courses on learning management systems," *Anat Sci Educ*, vol. 18, no. 9, pp. 1004–1012, Sep. 2025, doi: 10.1002/ase.70077.
- [11] C.-K. Looi *et al.*, *Methodologies and Intelligent Systems for Technology Enhanced Learning, 15th International Conference*. Springer Nature, 2025.
- [12] R. E. and C. Chakraborty, *Artificial Intelligence and IoT in Online Education Systems: Monitoring, Assessment, and Evaluation*. John Wiley & Sons, 2026.
- [13] Y. Wang, C. Wang, R. Li, and H. Lin, "On the use of BERT for Automated Essay Scoring: Joint learning of multi-scale essay representation," *arXiv [cs.CL]*, May 08, 2022.
- [14] F. F. Lubis *et al.*, "Automated Short-Answer Grading using Semantic Similarity based on Word Embedding," *IJTech*, vol. 12, no. 3, p. 571, Jul. 2021, doi: 10.14716/ijtech.v12i3.4651.
- [15] Y. Nie, "Automated essay scoring with SBERT embeddings and LSTM-Attention networks," *PeerJ Comput. Sci.*, vol. 11, p. e2634, Feb. 2025, doi: 10.7717/peerj-cs.2634.